

Harvesting AlphaEarth: Benchmarking the geospatial foundation model for agricultural downstream tasks

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ABSTRACT

Geospatial foundation models (GFM), pretrained on massive Earth observations (EO), have emerged as a promising approach to overcoming the limitations in existing featurization methods. Although most studies on GFMs have released the source codes and pre-trained weights, their deployment still demands extensive configuration, environment setup, inference EO preparation, and model fine-tuning. More recently, Google DeepMind has introduced AlphaEarth Foundation (AEF), a GFM pre-trained using multi-source EOs across continuous time. An annual and global embedding dataset is produced using AEF that is ready for analysis and modeling. The internal experiments show that AEF embeddings have outperformed operational models in 15 EO tasks without re-training. However, those experiments are mostly about land cover and land use classification. Applying AEF and other GFMs to agricultural monitoring requires an in-depth evaluation in critical agricultural downstream tasks. There is also a lack of comprehensive comparison between the AEF-based models and traditional remote sensing (RS)-based models under different scenarios, which could offer valuable guidance for researchers and practitioners. This study addresses some of these gaps by evaluating AEF embeddings in three agricultural downstream tasks in the U.S., including crop yield prediction, tillage mapping, and cover crop mapping. Datasets are compiled from both public and private sources to comprehensively evaluate AEF embeddings across tasks at different scales and locations, and RS-based models are trained as comparison models. AEF-based models generally exhibit strong performance on all tasks and are competitive with purpose-built RS-based models in yield prediction and county-level tillage mapping when trained on local data. However, we also find several limitations in current AEF embeddings, such as limited spatial transferability compared to RS-based models, low interpretability, and limited time sensitivity. These limitations suggest exercising caution when applying AEF embeddings in agriculture, where time sensitivity, generalizability, and interpretability is important. To our knowledge, this is the first study that systematically implements and evaluates embeddings from GFMs in agricultural downstream tasks across space, time, and spatial resolutions. The evaluation results and analyses can inform the design of future AEF versions and other GFMs and support their applications in agriculture and Earth science domains. Moreover, the proposed benchmarking workflow and datasets can be readily applied to evaluate future GFMs and facilitate their use in agricultural downstream applications.

1. Introduction

Earth observation (EO), grounded in remote sensing imagery, has enabled scalable and timely monitoring of dynamic Earth system processes, with agriculture being one of its most widely applied domains. Since the launch of the first Landsat satellite in 1972, there were early studies that explored the association between satellite spectral observations and crop conditions on the ground (Deering, 1978; Kauth and Thomas, 1976). Over the past half century, the rapid expansion of

satellite platforms and the availability of Earth observations have profoundly transformed research and practices in agroecosystems, such as cropland mapping (Zhang et al., 2025), crop yield prediction (Xiong et al., 2026), agricultural practice detection (Luo et al., 2023), agricultural carbon outcome measurement (Guan et al., 2023), and food security assessment (Nakalembe et al., 2021).

Despite the progress, there are still several bottlenecks that limit the application of EO in agriculture. A major challenge lies in the varying availability of EO data, driven by differences in satellite track overlap

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and cloud cover. Infrequent observations or data gaps can substantially reduce the effectiveness of cropland monitoring. Data harmonization is another common challenge when multi-modal EO data are used, arising from discrepancies in sensor design, spatial resolution, spectral response functions, and temporal revisit frequencies. In the past decade, machine learning (ML) and deep learning (DL) models based on EO have demonstrated potential across multiple agricultural downstream tasks. However, training ML/DL models requires abundant, high-quality, and task-specific labels. These ground truth labels, such as cropland types or crop yield records, are typically collected through expert annotation or field surveys, which are both time-consuming and resource intensive.

Recently, geospatial foundation models (GFM) have emerged as a promising approach to overcome the abovementioned bottlenecks, as they are trained on broad, large-scale data and can be readily adapted to a wide range of downstream EO tasks. GFMs are typically pre-trained in a self-supervised way, wherein the model exploits the input data itself to construct learning objectives, such as image reconstruction (He et al., 2022) or contrastive learning (Wang et al., 2022). This pre-training process requires no task-specific labels and enables the model to extract generic features from the input remote sensing data that can be adapted to diverse downstream tasks through fine-tuning or zero-shot learning.

Early GFMs were trained based on a single modality and targeted at a few related downstream tasks. For example, SatMAE (Cong et al., 2022), one of the first GFMs, was pre-trained on multi-spectral Sentinel-2 images using Masked Autoencoders (He et al., 2022) and evaluated on cropland classification and semantic segmentation. Similarly, ScaleMAE (Reed et al., 2023) was pre-trained by jointly learning representations from optical images at both low and high scales, achieving improved accuracy in 8 land-use classification datasets. In addition to encoding spatial information, NASA and IBM released a multi-temporal GFM named Prithvi-EO-2.0 (Szwarcman et al., 2024), which explicitly leverages transformer attention across both spatial and temporal dimensions via pre-training on a decade of satellite imagery from the Harmonized Landsat-Sentinel-2 archive. Recent efforts have shifted towards building multi-modal GFMs that accept inputs from more than one source, such as optical, multispectral, hyperspectral, and SAR images. For instance, the DOFA model (Xiong et al., 2024) incorporates a universal feature learning module designed for heterogeneous data modalities and was jointly trained using observations from five sensors. SkySense++ (Wu et al., 2025) integrates data from 11 satellite platforms and was pretrained progressively to learn both general representations and semantically enriched representations.

Although an increasing number of GFMs has been developed and released, several limitations remain in their application to practical downstream tasks. Even though most GFM studies have released source codes and pre-trained models, these models typically scale to millions or billions of parameters, resulting in significant operational overhead related to high-performance computing requirements (e.g., enterprise-grade GPUs). Meanwhile, adapting these GFMs requires fine-tuning using task-specific remote sensing observations, which still involve intensive image collection and pre-processing.

Most recently, Google DeepMind introduced AlphaEarth Foundation (AEF) (Brown et al., 2025), a GFM with ~ 480 M parameters that generates near-global geospatial representation that assimilates spatial, temporal, climate, topography, and measurement contexts across multiple sources. AEF is claimed to achieve continuous time-series EO featurization via several innovations (Brown et al., 2025), such as a Space-Time Precision encoder, an adaptive decoding scheme, and a spatially dense information time-bottleneck. Pre-trained using ~ 3 billion images sampled from ~ 5 million locations worldwide, AEF learns general representations that capture temporal dynamics of Earth's surface and climatic activities. At inference, it ingests annual multi-source EOs and encodes Earth's surface into 64-dimension embeddings at 10 m spatial resolution. Unlike traditional GFMs that require local deployment, annual AEF embeddings are publicly available as 'Analysis-Ready Data'

on Google Earth Engine in 2017–2025. It allows researchers to bypass heavy infrastructure requirements and focus on downstream task performance using standard ML libraries (e.g., Scikit-learn).

AEF can potentially transform EO tasks since it has largely addressed the challenges of data harmonization and data gaps. Internal evaluation within DeepMind showed that AEF embeddings have consistently outperformed previous featurization approaches tested on 15 EO tasks without re-training (Brown et al., 2025). However, certain aspects of AEF remain insufficiently evaluated. First, AEF embeddings have been evaluated mostly on land use and land cover mapping tasks, with no insights into critical agricultural downstream tasks, such as crop yield prediction and agricultural practice mapping. Second, DeepMind's internal evaluations were conducted in the same region and the same period, with no testing of the spatial or temporal generalizability of the AEF embeddings. More importantly, it remains unclear how AEF embeddings compare with commonly used remote sensing features across different scenarios, an evaluation that could offer valuable guidance for researchers and practitioners.

To answer these questions and facilitate better application of AEF embeddings in agriculture and potentially other EO tasks, we propose a benchmarking workflow that systematically evaluates AEF embeddings across three agricultural downstream tasks: crop yield prediction, tillage classification, and cover crop mapping. Ground truth labels were compiled from both public and private sources to comprehensively evaluate AEF embeddings across tasks at different scales and locations. In addition to local experiments, we also evaluated AEF's generalizability across space, time, and scales. Meanwhile, remote sensing-based models were trained as baselines for comparison in each task. Our contribution is twofold: to evaluate AEF for agricultural tasks beyond land cover classification, and to demonstrate the importance of testing GFMs across regions, years, and scales, since these are three critical axes of generalization for real-world Earth science applications. Generalizability across regions, years, and spatial scales is critical because remote sensing models are not used only under the conditions in which they are trained. In practice, models calibrated on historical data are applied to future years for forecasting, adapted from data-rich to data-scarce regions, and deployed at spatial scales different from the training data (e.g., trained on county-level yields but applied to fine-scale grids). Evaluating robustness across space, time, and scale therefore reflects how these models are used in operational agricultural monitoring. Thus, our work provides a methodological blueprint for future GFM evaluation in real-world operational settings.

The structure of the paper is as follows: In Section 2, we describe each of the downstream task and the corresponding sources of the ground truth data. In Section 3, we introduce the AEF embeddings and the remote sensing features. The details about the data downloading and pre-processing procedures are provided. In addition, ML models and evaluation schemes are illustrated. In Section 4, we present the evaluation results of the AEF-based models and RS-based models across three tasks. The space- and scale-transfer experiments have also been conducted. In Section 5, we summarize the advantages and limitations of AEF embeddings in agricultural applications. In Section 6, we conclude this study and discuss potential future research.

2. Experimental tasks and ground truth data

We evaluated the potential of AEF embeddings for crop yield prediction, tillage mapping (tillage type classification), and cover crop mapping (cover crop detection) at both the regional scale (county level) and the field scale. The corresponding agricultural data were collected as ground truth labels for model training and evaluation. In particular, the county-level agricultural data are provided by the governments' agricultural agencies, which are publicly available. Field-level agricultural data are typically stored in governmental databases or commercial systems and are hard to acquire due to privacy concerns (Deines et al., 2021; Ma et al., 2024b). Corteva Agriscience collected and provided the

agricultural data at the field level in the U.S. for model training and evaluation in this study. More details are given below.

2.1. Crop yield prediction

Crop yield prediction provides essential information for farm resource management, food security monitoring, and market planning (Becker-Reshef et al., 2019). Many governments conduct monthly and annual agricultural surveys to estimate crop progress and predict yield within the growing season. More recently, ML-based yield prediction models using EO have proven an effective and cost-efficient approach that can accurately predict crop yields. Specifically, raw remote sensing or derived vegetation indices are used as predictor variables to train ML models, together with the corresponding ground truth yield records. The trained ML models are implemented to predict yields in unseen regions or years.

We evaluated the performance of AEF embeddings for yield prediction in the U.S., which is the world's largest agricultural producer and export (Kogan et al., 2013). The county-level yield data for corn, soybean, and winter wheat in the U.S. from 2017 to 2024 were obtained from the United States Department of Agriculture National Agricultural Statistics Service (USDA-NASS) (USDA-NASS, 2024), which publishes annual yield statistics after the growing season. Corn and soybean yield data were collected from 12 U.S. Corn Belt states and winter wheat yield data from 11 Midwestern states, which are the major crop producers and account for more than 70% of the national crop production (Fig. 1). We converted the unit of yield from bushels per acre (bu/ac) to tonnes per hectare (t/ha).

In addition, we also evaluated AEF embeddings on field-level yield prediction, which provides fine-scale information to support farm management. The ground-based measures of yields were collected using in-field yield monitoring system during harvesting by Corteva Agriscience (Deines et al., 2021). This Corteva yield datasets include multi-year field-level yield data and the field locations for corn, soybean, and winter wheat. Corn and soybean fields distribute across the U.S. Corn Belt and winter wheat fields span the U.S. Wheat Belt and Midwest (Ma et al., 2024b), including 89,938 corn fields and 73,492 soybean fields in 2017–2018, and 20,401 winter wheat fields in 2017–2022.

2.2. Spring tillage mapping

Tillage is an agricultural practice that farmers implement to prepare the land for sowing. Conventional high-intensity tillage often leaves little crop residue and causes large disturbance to the soil, which accelerates soil erosion and increase greenhouse gas emissions (Lu et al., 2022). Low-intensity tillage, including conservation tillage and no-till,

has been promoted worldwide to conserve soil and water (Claassen et al., 2017). Due to its far-reaching environmental impacts, tillage mapping based on ML and remote sensing observations has drawn considerable research attention. In addition to raw spectral bands and VIs, tillage-related indices have been derived to detect crop residue cover and classify tillage practices, such as Normalized Difference Tillage Index (NDTI) (Van Deventer et al., 1997).

We evaluated the performance of AEF embeddings for spring tillage mapping in the U.S. Corn Belt at both the county level and the field level (Fig. 1 (a)). County-level tillage statistics were collected from USDA-NASS, which record the area of cropland managed under different tillage systems, including conventional tillage, conservation tillage, and no-till. The tillage statistics are compiled through the USDA Census of Agriculture every five years and are currently available in 2017 and 2022. We collected the data and combined conservation tillage and no-till as low-intensity tillage. The proportion of cropland under low-intensity tillage within each county were calculated as the ground truth data, and there are overall 2,001 county-year records.

Field-level spring tillage data were collected from farmers by Corteva Agriscience. The Corteva tillage dataset contains grower-submitted records of tillage operations and the field boundaries spanning 2016–2023 throughout the U.S. Corn Belt. We used the data in 2017–2023 and categorized the tillage activity for each field into low- or high-intensity tillage according to the corresponding intensity levels, resulting in a total of 24,514 field-year records.

2.3. Cover crop mapping

Cover cropping is another conservation agricultural practice in which non-cash crops are cultivated during the interval between the harvest of cash crops and the planting of the next (Plastina et al., 2020). It helps prevent periods of bare soil in croplands and reduce the risk of soil erosion (Koudahe et al., 2022), nitrogen leaching (Abdalla et al., 2019), and weed infestation (Alonso-Ayuso et al., 2018). In the U.S., the adoption of cover crops has expanded substantially in recent decade, with the national cover crop area nearly doubling from 10.3 million acres in 2012 to 18.0 million acres in 2022 (Lobell et al., 2025; Zulauf et al., 2024). EO-based methods have been developed to detect and track the adoption of cover crops (Fendrich et al., 2023; Zhou et al., 2022).

We evaluated the performance of AEF embeddings for cover crop mapping in the U.S. Corn Belt at the field level (Fig. 1 (a)). Again, the field-level cover crop data were collected from farmers by Corteva Agriscience. Each data record specified if a cover crop was used in a field and provided the field boundary, the start and end dates of cover cropping, and the species planted. The annual cover crop dataset covers the 12 states in the U.S. Corn Belt and spans from 2017 to 2024. Since

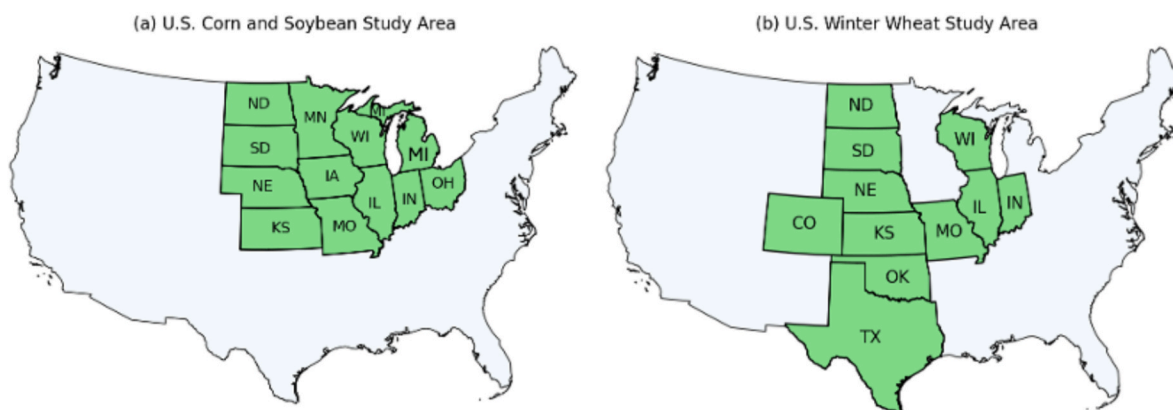


Fig. 1. The study areas in the U.S., covering (a) 12 states in the U.S. Corn belt for corn and soybean, and (b) 11 Midwestern states for winter wheat. The states include Colorado (CO), Illinois (IL), Indiana (IN), Iowa (IA), Kansas (KS), Michigan (MI), Minnesota (MN), Missouri (MO), Nebraska (NE), North Dakota (ND), Oklahoma (OK), Texas (TX), South Dakota (SD), and Wisconsin (WI).

the planting dates of the cover crops can happen in the fall after the harvest, or in the spring before the planting, we concatenated two-year AEF embeddings as the input predictors. Therefore, we dropped year 2017 and did experiments for 2018–2024, leading to a total number of 47,709 field-year samples. We did not conduct county-level cover crop mapping experiments because county-level data are available only for 2017 and 2022, and AEF embeddings for 2016 are unavailable, which prevents the construction of two-year AEF embedding inputs as predictors.

A summary of the datasets and experimental settings for each downstream tasks is given in Table 1.

3. Materials

3.1. Alpha Earth Foundation model

The AEF model was trained on ~ 3 billions of observations across optical (Landsat 8/9, Sentinel-2), radar (Sentinel-1), LiDAR (GEDI), Climate (ERA5-Land), gravity fields (GRACE), Elevation (GLO-30), and text sources (Wikipedia) (Brown et al., 2025). A space–time encoder and a teacher–student framework was employed to capture spatial, temporal, and measurement contexts in a compact form. Consequently, it transformed sparse, heterogeneous EOs from multiple sources into a universal 64-dimensional embedding space for global mapping. The resulting 64-dimensional embeddings are time-continuous and sensor-agnostic at 10-meter resolution, achieving near-global spatial coverage and excluding polar extremes beyond $\sim 82^\circ$ N/S.

The AEF embeddings are publicly available as annualized, global layers spanning 2017 to 2024, hosted on Google Earth Engine (GEE) as an image dataset. The 64-band annual embeddings are stored as image layers, with individual bands labeled sequentially from A00 to A63.

3.2. Remote sensing, climate, and topographic data

In addition to AEF embeddings, we extracted traditionally used features from remote sensing, climate, and topographic sources, which are available on GEE.

3.2.1. Satellite optical remote sensing

The Landsat family of sensors was selected, which provide 30-meter spatial resolution global observations over decadal time spans. All available Landsat Collection 2 Tier 1 Land Surface Reflectance data from 2017 to 2024 were used, including Landsat 7 Enhanced Thematic Mapper Plus (ETM+), Landsat 8 Optical Land Imager (OLI), and Landsat 9 OLI. We extracted the reflectance data from six bands, including Red, Green, Blue, Near Infrared (NIR), Shortwave Infrared 1 (SWIR1), and Shortwave Infrared 2 (SWIR2).

3.2.2. Climate reanalysis and topographic data

Climate features are widely used in agricultural modeling as they capture environmental drivers of agroecosystem dynamics and complement remote sensing features. We chose the ERA5-Land Climate Reanalysis dataset (Muñoz-Sabater et al., 2021), which provides daily

global data at 10 km resolution by integrating land surface modeling with in-situ observations worldwide. From this dataset, we extracted precipitation (PPT), maximum temperature, and minimum temperature. They are used to calculate the accumulative PPT and the growing degree days (GDD) in each month of the growing season:

$$PPT_j = \sum_{d=1}^M PPT_d \quad (1)$$

$$GDD_j = \sum_{d=1}^M \sum_{h=1}^{24} (\max(0, \min(T_h - T_{\min}, T_{\max} - T_{\min}))) \quad (2)$$

in which M denotes the number of days in the month j , PPT_d is the daily total precipitation (in mm) (Eq. (1)), T_h is the hourly temperature (in $^\circ\text{C}$) and calculated based on sinusoidal interpolation between the daily minimum and maximum temperatures (Eq. (2)), and $T_{\max}$ ($T_{\min}$) denotes the upper (lower) biological temperature threshold, beyond which crop development no longer respond to temperature variations, which is 30°C (8°C) for soybean and 26°C (0°C) for winter wheat (Ritchie, 1991; Swan et al., 1987).

Moreover, topographic characteristics play a significant role in agricultural decision-making and have been widely utilized in tillage classification. We extracted Elevation data from the USGS 3D Elevation Program 10 m National Map (Archuleta et al., 2017) to describe the topographic characteristics in the study area.

3.3. Data cleaning and downloading

We used GEE to collect and download each dataset. Before downloading the data, noisy pixels in satellite imagery were first removed by applying the per-pixel quality mask (clouds, cloud shadows, snow/ice, and water). Following that, we used the 30-meter U.S. cropland layer (CDL) (Boryan et al., 2011) to keep pixels on specific crop types and mask out non-cropland pixels. Next, observations on the remaining pixels were aggregated to the county (field) level by calculating the mean values within each county (field) boundary. No additional resampling was applied during this process. Finally, the aggregated observations were downloaded from GEE, including AEF embeddings, time-series satellite remote sensing data, monthly climate features, and elevation.

3.4. Remote sensing data processing

The frequency of time-series remote sensing observations varies spatially due to different satellite path overlap among adjacent tracks and cloud coverage. Therefore, it is challenging to use raw time-series observations as predictors. Instead, we pre-processed the time-series remote sensing data and generated predictor variables tailored to each task.

3.4.1. Features for yield prediction

In yield prediction tasks, we downloaded the time-series satellite imagery across the whole growing season for each crop. In addition to six spectral bands, two vegetation indices (VIs), Normalized Difference Vegetation Index (NDVI) and Green Chlorophyll Vegetation Index (GCVI), were calculated (Eq. (3)–(4)). NDVI is a classic remote sensing index that measures vegetation vigor and greenness by contrasting NIR and Red reflectance. GCVI extends this approach by incorporating green reflectance, making it more directly sensitive to chlorophyll content and photosynthetic capacity while partially alleviating NDVI's saturation issues in high-biomass regions. (Gitelson et al., 2003).

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (3)$$

$$GCVI = \frac{NIR}{Green} \quad (4)$$

Table 1

A summary of the experimental settings.

Task	Level	Year	Crop	# Samples
Yield Prediction	County	2017–2024	Corn	6,325
			Soybean	6,024
			Winter Wheat	3,020
	Field	2017–2018	Corn	89,938
			Soybean	73,492
			Winter Wheat	20,401
Tillage Mapping	County	2017&2022	N/A	2,001
	Field	2017–2023	N/A	24,514
Cover Crop Mapping	Field	2018–2024	N/A	47,709

We fit a second-order harmonic regression (Wilson et al., 2018) to each spectral bands and VIs based on all available observations during the growing season (Eq. (5)).

$$y(t) = c + a_1 \cos(2\pi t) + b_1 \sin(2\pi t) + a_2 \cos(4\pi t) + b_2 \sin(4\pi t) \quad (5)$$

where t represents the date of the observation; c denotes the intercept coefficient; a_1 and b_1 (a_2 and b_2) represent the first-order (second-order) cosine and sine coefficients, respectively.

The five harmonic coefficients (a_1 , b_1 , a_2 , b_2 , c) for spectral bands and VIs, which summarize the time-series satellite data, were used as predictor variables. In addition, we extracted phenology-based metrics from the fitted harmonic curves. For each spectral and VI bands, the maximum value in the growing season was picked and termed as Band_peak. In addition, band values were derived for the periods 30 days before and after the peak date, referred to as Band_b30 and Band_a30. Moreover, the partial integral approach was used to compute the area under the curve between these three time points (Deines et al., 2021). The area under the curve between Band_peak and Band_b30 was termed as Band_b30_int. The area under the curve between Band_peak and VI_a30 was termed as VI_a30_int. These indicators capture vegetation status during the phases leading up to and following the peak, offering information on growth trajectories. An example of the time-series NDVI and the derived harmonic features is present in Fig. 2.

In total, 80 harmonic features were generated for each county or field, comprising 10 features for each of the 6 spectral bands and 2 VIs. Meanwhile, monthly GDD and PPT were used as predictor variables in the U.S., which added 10 additional predictors for corn and soybean (May to Sep) and 12 additional predictors for winter wheat (Jan to Jun).

3.4.2. Features for tillage mapping

For tillage mapping, we further calculated three tillage indices that quantify crop residue cover, including NDTI, Simple Tillage Index (STI), and Crop Residue Cover index (CRC) (Eq. (6)–(8)) (Eskandari et al., 2016; Sullivan et al., 2006; Van Deventer et al., 1997):

$$\text{NDTI} = \frac{\text{SWIR1} - \text{SWIR2}}{\text{SWIR1} + \text{SWIR2}} \quad (6)$$

$$\text{STI} = \frac{\text{SWIR1}}{\text{SWIR2}} \quad (7)$$

$$\text{CRC} = \frac{\text{SWIR1} - \text{Blue}}{\text{SWIR1} + \text{Blue}} \quad (8)$$

The tillage mapping experiments were conducted in the U.S. Corn Belt, where the cropping systems are mainly corn and soybean, and most tillage is implemented in spring (Lu et al., 2022). We organized monthly remote sensing features by extracting the monthly minimum and maximum values from times-series observations in April to June for each band.

A total of 66 monthly remote sensing features was used for each county or field, comprising 2 features for each of the 6 raw bands, 2 VIs (GCVI and NDVI), and 3 tillage indices (NDTI, STI, CRC) in each month. In addition, elevation was included as a predictor variable to describe the terrain characteristics.

3.4.3. Features for cover crop mapping

Cover crops can be planted and terminated at any point after the harvest and before planting in the next season, a period that spans from October to May of the next year in the U.S. Corn Belt. For each field in the Corteva cover crop dataset, we collected the time-series Landsat six raw bands and two VIs (NDVI and GCVI) during this period and fitted a second-order harmonic regression. Following that, we extracted the minimum and maximum values for each band from the fitted curves within each month from October to May. Meanwhile, we calculated monthly mean temperature (Tmean) and accumulative PPT for each field using the ERA5-Land Climate Reanalysis dataset. The extracted predictors were paired with the corresponding cover crop labels (Class 0: no cover crop; Class 1: cover crop) for model training and evaluation. The total number of predictors is 144 for each field, including 18 monthly features across 8 months.

3.5. Evaluation models and schemes

Two widely used ML models for tabular data were trained and evaluated for each task, including RF and XGBoosting (XGB) (Breiman, 2001). Both models were implemented using Scikit-learn with 200 individual trees (Pedregosa et al., 2011). We did not select DL models because the limited number of training samples in certain tasks would likely be insufficient to train reliable DL models. Moreover, RF and XGB have shown to outperform, or perform comparably to DL models on tabular data, while requiring substantially less computational resources (Hollmann et al., 2025). In addition, we also trained baseline models based on traditional remote sensing feature sets. Each downstream task required different predictor variables (Section 3.4), which we collectively refer to as remote sensing (RS) features. Each RF and XGB model was trained separately using AEF embeddings and RS features for each

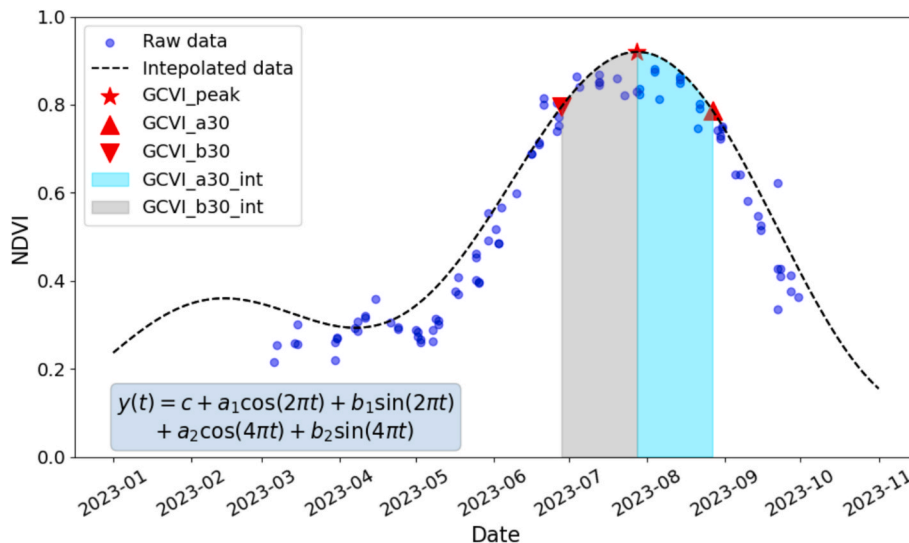


Fig. 2. An example of the 2023 time-series NDVI and the derived harmonic features for corn lands in Adams County, Illinois, USA.

task (Table 1). We note that DL models based on raw time-series RS data can potentially be stronger baselines. However, this study aims to develop an evaluation workflow that is easily reproducible by both DL specialists and practitioners in agriculture and Earth observation. Tree-based models typically require fewer hyperparameter choices and less computational tuning than DL models, making them more practical for transparent and repeatable benchmarking.

A standard scheme to evaluate ML model performance is random cross validation (CV), in which the dataset is randomly split to a training set and a test set. In geospatial tasks, random CV can lead to inflated model performance due to spatial autocorrelation, as nearby counties/fields with similar properties may be split between the training and testing sets. Instead, we evaluated the model performance in each task using two schemes: spatial-temporal CV and yearly CV. In spatial-temporal CV, samples from the same region and years are grouped into either the training or the testing set, avoiding spatial autocorrelation. Specifically, we implemented State-Year CV for county-level data, in which county samples from different state-year groups are used for training and testing, separately. Similarly, for the field-level data, we implemented County-Year CV. In addition, yearly CV, also known as leave-one-year-out evaluation, reflects a more practical scenario, when data from a single year serve as the test set and data from the remaining years are used for training.

In addition, we designed scale-transfer and space-transfer experiments to assess whether embeddings enhance accuracy and generalizability across scale and spatial domains. In scale-transfer experiments, ML models for yield prediction are trained using county-level yield data and applied to predict yield at the field level. In space-transfer experiments, ML models are trained and tested across two ecoregions in the U. S. Corn Belt. Specifically, the study counties were grouped into two ecoregions as defined by the United States Environmental Protection Agency (EPA): the Eastern Temperate Forests (ETF) and the Great Plains (GP). The ETF is characterized by a warm, humid, temperate climate, with humid summers and mild to cold winters. In contrast, the GP consists mainly of flat grasslands with limited forest cover, featuring very hot summers and harsh winters (Ma et al., 2021). The ETF mainly includes five eastern states (Fig. 1 (a)), including Illinois, Indiana, Michigan, Ohio, and Wisconsin. The GP mainly covers seven states on the western side of the Corn Belt (Fig. 1 (a)), including Iowa, Kansas, Minnesota, Missouri, North Dakota, Nebraska, and South Dakota. For simplicity, we termed ETF and GP as East and West, respectively.

We have summarized the evaluation scheme used in this study in Table 2. This evaluation scheme is built on four pillars: (1) assessing local performance using standard spatial-temporal cross-validation; (2) testing temporal generalizability with a yearly CV scheme; (3) testing spatial transferability across distinct ecoregions; and (4) evaluating scale transferability from coarse to fine resolutions. This holistic approach allows us to diagnose not just how well the embeddings perform, but under what conditions they succeed or fail, providing generalizable insights for the broader GFM field.

Each experiment is repeated five times under different random seeds, and the mean evaluation results are presented. For regression tasks,

coefficient of determination (R^2) and root mean squared error (RMSE) are calculated as the evaluation metrics (Eq. (9) – (10)). For classification tasks, the evaluation metrics include the overall accuracy (Accuracy), F1 scores for each class (F1-0 and F1-1), and the average F1 score weighted by the number of true instances (F1-weighted) (Eq. (11) – (14)):

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (9)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (10)$$

$$Accuracy = \frac{TP + TN}{N} \quad (11)$$

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (14)$$

where N denotes the total number of samples in the evaluation set; y_i is the ground truth data and $\bar{y}$ is the average value of the ground truth data; $\hat{y}_i$ is the predicted values by the models. TP , TN , and FP represent the number of True Positive, True Negative, and False Positive classifications by the models.

4. Experiment results

4.1. Crop yield prediction

4.1.1. County-level evaluation

In the U.S., the AEF-based models had comparable performance as the RS-based models under the State-Year CV, achieving an R^2 of around 0.80 for corn and soybean, and 0.70 for winter wheat (Table 3). All models showed a slight reduction in accuracy under the yearly CV scheme, yet AEF-based models consistently outperformed RS-based models, highlighting the stronger temporal generalizability of AEF embeddings.

Scatter plots of the ground truth and predicted yields (Fig. 3) indicate that both AEF-based and RS-based models align well with the reported yields from USDA-NASS. Considering that the XGB models slightly outperform the RF models in most cases, the comparison is presented using the results from the XGB models. RS-based models showed notable underestimation of winter wheat yields in high-yield ranges (Fig. 3(c2)), primarily due to the limited number of high-yield samples in the training set. AEF-based models partially mitigated biases in high-yield ranges and achieved higher accuracy (Fig. 3(c1)).

4.1.2. Field-level evaluation

The performance of AEF embeddings was further evaluated for field-level yield prediction based on the Corteva dataset of over 180,000 yield records (Table 1). Predicting field-level yields is typically more challenging than at the county level, primarily due to noisier yield records, reduced satellite data coverage, and the mismatch between fine-scale fields and the coarse spatial resolution of climate variables. Therefore, we observed smaller R^2 and larger RMSE in the field-level evaluation results (Table 4). Similar to the county-level results, AEF-based models achieved comparable performance as the RS-based models under the county-year CV and the yearly CV (Table 4).

We further compared the model performance under the scale-transfer scheme, in which the models were trained using predictors and yield records at the county level and implemented to predict yields

Table 2

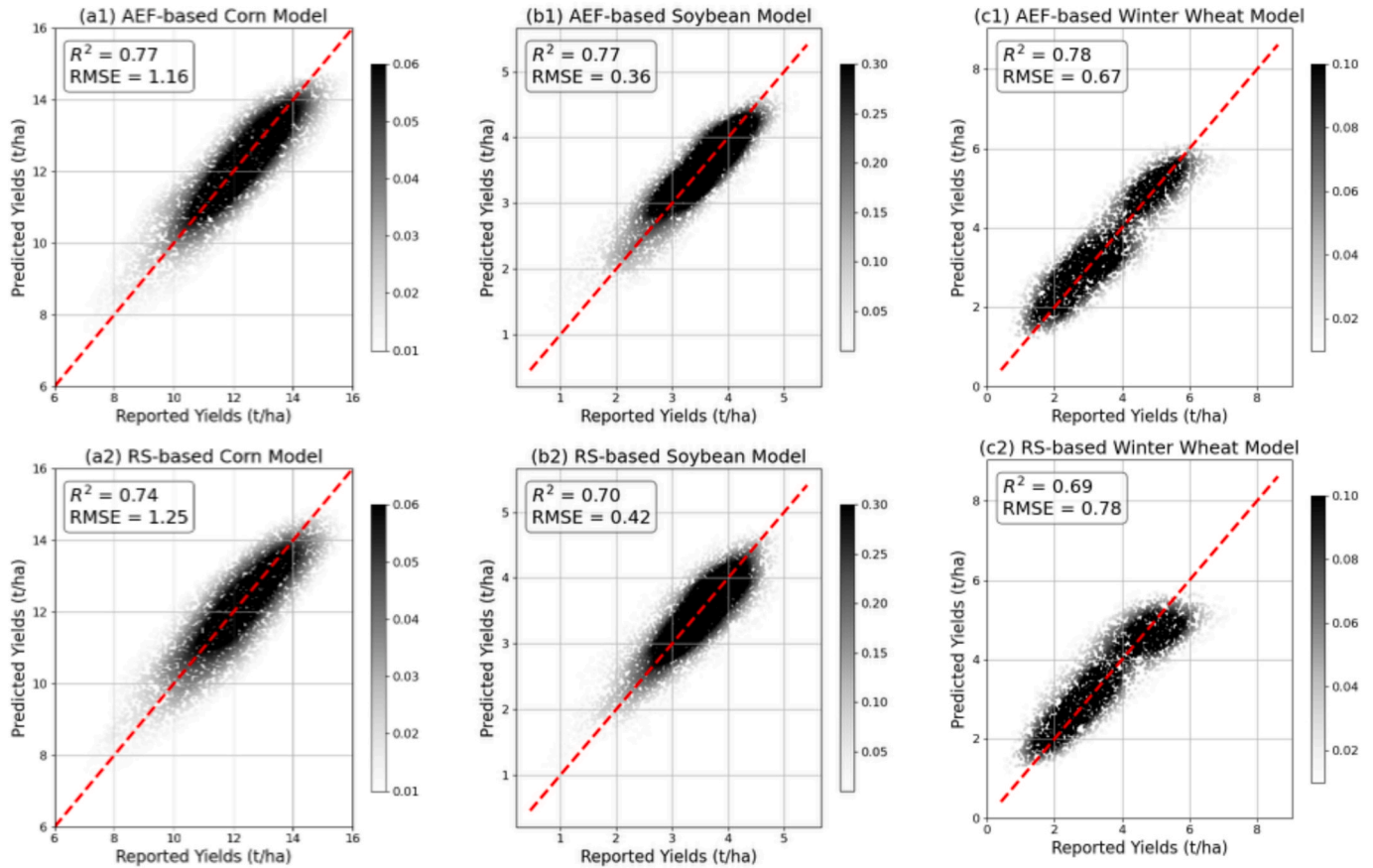
An overview of the evaluation schemes in this study.

Scheme	Training Set	Test Set	Objective
State (County)-Year CV	Data in specific state (county)-year groups	Data in the rest state(county)-year groups	Avoid spatial and temporal autocorrelation
Yearly CV	Data in all but one years	Data in the remaining year	A practical scenario where training data are available in certain years
Scale-Transfer	County data	Field data	Evaluate the scale transferability
Space-Transfer	Data in one ecoregion	Data in the other ecoregion	Evaluate the spatial transferability

Table 3

Evaluation results of county-level crop yield prediction in the U.S. in 2017–2024. The best performer is highlighted in bold for each case.

Crop	Scheme	AEF				RS			
		RF		XGB		RF		XGB	
		R ²	RMSE	R ²	RMSE	R ²	RMSE	R ²	RMSE
Corn	State-Year	0.78	1.19	0.78	1.18	0.79	1.14	0.80	1.12
	Yearly	0.77	1.15	0.77	1.16	0.76	1.20	0.74	1.25
Soybean	State-Year	0.75	0.38	0.78	0.36	0.78	0.37	0.79	0.36
	Yearly	0.74	0.39	0.77	0.36	0.71	0.41	0.70	0.42
Winter Wheat	State-Year	0.76	0.68	0.78	0.64	0.70	0.78	0.71	0.76
	Yearly	0.77	0.68	0.78	0.67	0.69	0.79	0.69	0.78

**Fig. 3.** The scatter plots of yield prediction results by XGB under the yearly CV in the U.S. for (a) Corn, (b) Soybean, and (c) Winter Wheat in 2017–2024.**Table 4**

Evaluation results of field-level crop yield prediction in the U.S. The best performer is highlighted in bold for each case.

Crop	Scheme	AEF				RS			
		RF		XGB		RF		XGB	
		R ²	RMSE	R ²	RMSE	R ²	RMSE	R ²	RMSE
Corn	County-Year	0.43	2.28	0.44	2.27	0.45	2.24	0.46	2.24
	Yearly	0.31	2.54	0.30	2.55	0.33	2.50	0.33	2.49
	Scale-Transfer	0.24	2.67	0.25	2.64	0.37	2.43	0.38	2.40
Soybean	County-Year	0.36	0.87	0.37	0.87	0.35	0.90	0.35	0.90
	Yearly	0.29	0.92	0.31	0.91	0.28	0.93	0.25	0.95
	Scale-Transfer	0.13	1.02	0.11	1.03	0.22	0.97	0.24	0.95
Winter Wheat	County-Year	0.43	1.32	0.43	1.31	0.42	1.30	0.41	1.31
	Yearly	0.38	1.34	0.40	1.32	0.36	1.35	0.36	1.36
	Scale-Transfer	−0.18	1.85	−0.04	1.73	0.20	1.52	0.22	1.50

at the field level (Ma et al., 2024b). This scheme evaluates predictors' transferability across spatial scales. AEF-based models significantly

underperformed the RS-based models in this task (Table 4). This is likely due to the compression inherent in embedding representations, which,

when aggregated to the county level, reduces the model's sensitivity to subtle yield-driving factors at the field scale. For instance, each single embedding may capture different signals across locations within a county. By averaging all AEF pixels to generate county-level representations, field-level information relevant to crop growth can be diluted or lost, leading to weaker predictive power at the field level. In contrast, remote sensing features preserve more direct physical signals (e.g., canopy structure, vegetation vigor, and soil moisture) that are universally interpretable across scales, enabling stronger generalization across scales.

4.2. Tillage classification

4.2.1. County-level tillage mapping

For county-level tillage mapping, ML models were trained to estimate the ratio of cropland under low-intensity tillage in each county. When evaluated under the state-year CV, AEF-based and RS-based models provided broadly comparable predictive skill, with R^2 values around 0.48 and RMSE near 12.20% (Table 5). When assessed under yearly CV, the AEF-based RF model achieved the strongest performance in 2017 ($R^2 = 0.56$, RMSE = 11.85%), whereas for 2022 the advantage shifted toward RS-based RF model ($R^2 = 0.53$, RMSE = 12.16%). When aggregating across all years, these differences tend to smooth out, resulting in modestly higher R^2 for AEF-based models (0.52) compared to RS-based models (0.50), though overall accuracies remain similar (Table 5).

4.2.2. Field-level tillage mapping

For field-level tillage mapping, ML models were trained to classify the tillage activity of each field into high-intensity tillage (class 0) or low-intensity tillage (class 1). Under the county-year CV, RS-based models achieved slightly better performance than AEF-based models, while all models had comparable predictive power, with accuracy and weighted F1 scores around 0.75–0.80 (Fig. 4). High-intensity tillage (class 0) is more difficult to detect because it requires remote sensing observations during the brief period when crop residues are removed and major field disturbances occur. Consequently, RS-based models yielded lower F1-0 scores compared to F1-1, and AEF-based models exhibited the same pattern (Fig. 4).

In particular, large variations were observed in the yearly CV since the timing and intensity of tillage activities vary from year to year. RS-based models attained accuracy and weighted F1 scores around 0.75, while the scores of AEF-based models dropped to around 0.70 (Fig. 5). AEF-based models had lower F1-0 scores with considerable variability. These results indicate that AEF embeddings are not inherently superior to RF features for detecting changes that occur within short time windows.

4.3. Cover crop mapping

For field-level cover crop mapping, ML models were trained to classify fields as either without (class 0) or with cover crops (class 1). Generally, RF models had better performance than XGB models. The degraded performance by XGB is potentially due to the noise in the input feature sets, since there were fewer satellite observations during winter

and early spring because of high cloud coverage and the presence of snow and ice. (Fig. S1), leading to more noise in the remote sensing predictors. Prior work has similarly found that RF is more robust to noisy observations than XGB (Fawagreh et al., 2014).

Under both schemes, RS-based RF models had the highest overall performance, with accuracy and weighted F1 scores around 0.80–0.85 (Figs. 6–7). AEF-based models performed well in detecting cover crops (higher F1-1) but misclassified more negative samples. Again, under the yearly CV (Fig. 7), all models showed large variations in performance, likely reflecting the varying timing of cover cropping from year to year. AEF-based models exhibited the largest variations (Fig. 7), reflecting the disproportional representation by AEF embeddings across time.

4.4. Space-transfer evaluation

Remote sensing spectral features and climate variables generally exhibit significant geographic shifts due to variations in factors such as crop planting and harvesting dates and soil types across different regions (Ma et al., 2024a; Tong and Wang, 2025). AEF was trained with more than 3 billion images worldwide, with the potential to learn spatial-invariant features. We assessed the spatial transferability of AEF embeddings for each of the downstream tasks across an eastern ecoregion ETF (East) and a western ecoregion GP (West) within the U.S. Corn Belt at both the county and the field levels. AEF-based ML models were trained to predict yields for corn and soybean in one ecoregion and then were tested in the other, and vice versa (Section 3.5). We also trained and tested the RS-based models for comparison. Winter wheat yield prediction tasks were excluded since the wheat data were primarily located in the East. In the yield prediction experiments, we only present the results by RF models since RF and XGB had very similar performance. In the field-level tillage and cover crop mapping experiments, we also only present the RF model's results since it has proven more robust in the previous experiments.

In the yield prediction tasks, the AEF-based models had substantially worse space-transfer performance than the RS-based models. In particular, when trained in the East and tested in the West (East → West), AEF-based models failed to make accurate yield prediction for either corn or soybean in the West, with R^2 values around -0.28 – 0.07 (Table 6). When trained in the West and tested in the East (West → East), the AEF-based models had improved performance in the East, with R^2 values around 0.39–0.51, possibly due to having more states and thus more representative training samples in the West (Table 6). Meanwhile, RS-based models had stable performance under both space-transfer schemes and consistently outperformed AEF-based models (Table 6). Similar patterns were observed at the field level (Table 6).

We observed mixed patterns in the tillage mapping and cover crop mapping tasks (Table 6). In particular, in county-level tillage mapping, AEF-based models performed well when transferring from West to East, but underperformed RS-based models when transferring from East to West. At the field level, RS-based RF achieved higher accuracy under both schemes but with small margins. Similarly, in field-level cover crop mapping, AEF-based RF and RS-based RF exhibited very similar spatial transferability. One possible reason is that agricultural practices are less influenced by differences among ecoregions than crop yields.

Table 5

Evaluation results of county-level low-intensity tillage mapping in the U.S. The best performer is highlighted in bold for each case.

Scheme	Year	AEF				RS			
		RF		XGB		RF		XGB	
		R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE
State-Year	2017&2022	0.47	12.22%	0.47	12.22%	0.48	12.18%	0.47	12.32%
Yearly	2017	0.56	11.85%	0.54	12.04%	0.46	13.00%	0.48	12.79%
	2022	0.47	12.91%	0.42	13.59%	0.53	12.16%	0.48	12.84%
	All	0.52	12.40%	0.48	12.85%	0.50	12.61%	0.48	12.82%

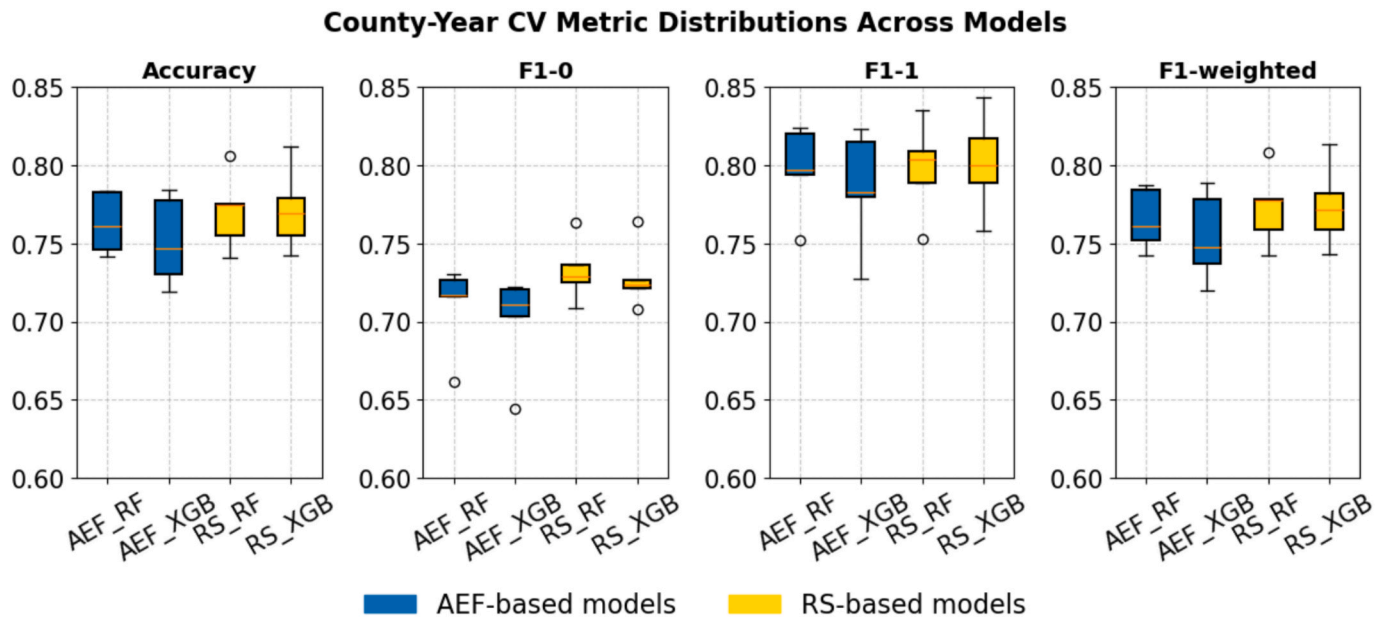


Fig. 4. The field-level tillage mapping model performance (accuracy, F1-0, F1-1, and F1-weighted) under the county-year CV scheme. Each experiment is repeated five times, and each box contains the scores of all iterations.

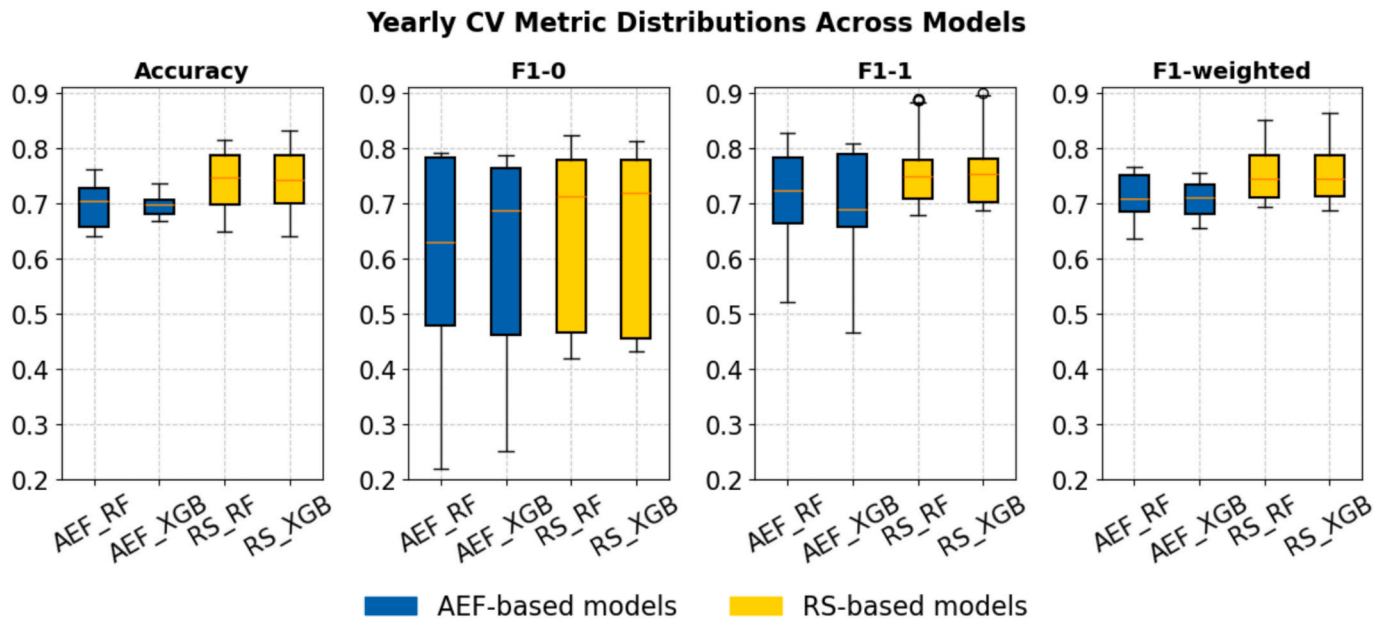


Fig. 5. The field-level tillage mapping model performance (accuracy, F1-0, F1-1, and F1-weighted) under the yearly CV scheme. Each experiment is repeated five times, and each box contains the scores of all iterations.

5. Discussion

5.1. Advantages of AEF embeddings

Our experiments demonstrated several advantages of AEF embeddings over commonly used RS features:

Automatic data harmonization: AEF embeddings inherently harmonize data from multiple sources, including optical, multispectral, radar, LiDAR, gravity fields, climate, topographic, and text sources. AEF reconciles multiple sparse, non-uniformly observation records with varying resolutions and formats into a continuous record with a 10-meter spatial resolution. Ablation experiments in the AEF paper showed that each modality has contributed positively and increased the

accuracy in downstream tasks (Fig. 4 in (Brown et al., 2025)). As such, users do not need to collect data from each individual source or design feature engineering strategies to harmonize them.

Fast data downloading and processing: The annual AEF embeddings are stored as an image dataset on GEE, and each year's data are organized as a raster image. This data format allows fast data downloading from GEE. Also, the AEF embeddings require no pre-processing and are ready for modeling. In contrast, downloading RS imagery from GEE takes much more time, as it involves extensive data filtering, cloud masking, and spatial aggregation on each observation date. The downloaded time-series RS images require intensive pre-processing and quality control before they can be used for model training and inference.

Competitive performance in local experiments: AEF-based

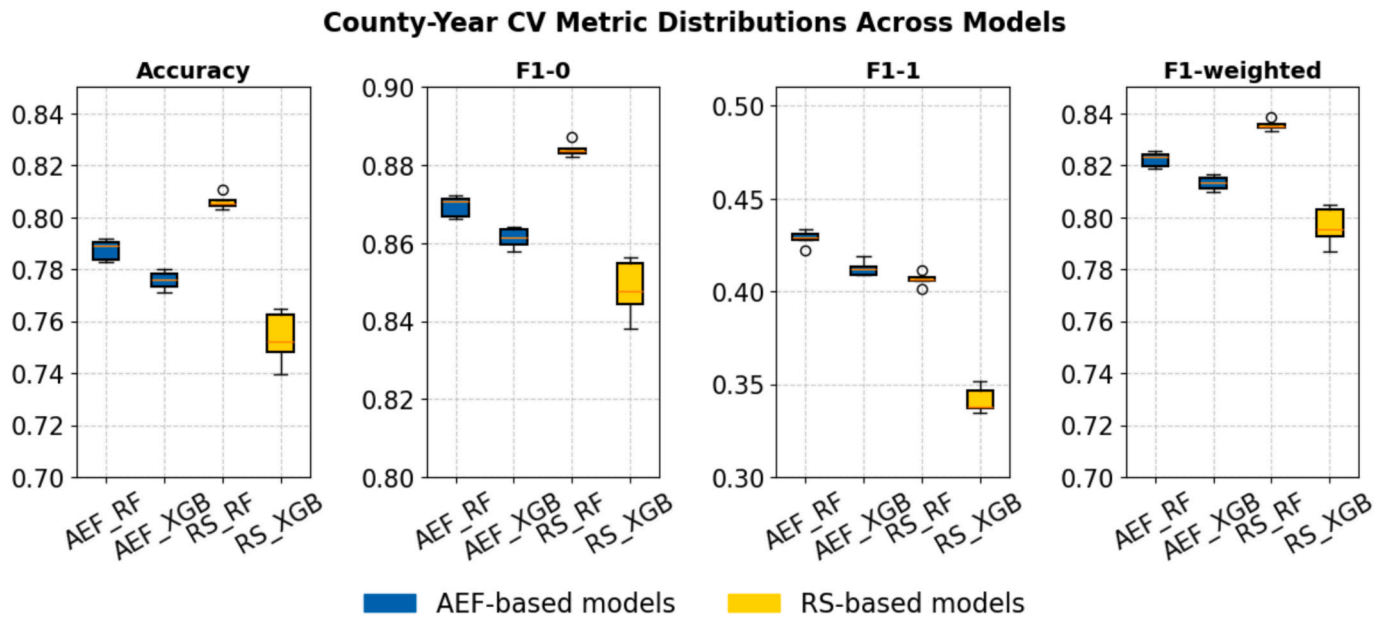


Fig. 6. The field-level cover crop mapping model performance (accuracy, F1-0, F1-1, and F1-weighted) under the county-year CV scheme. Each experiment is repeated five times, and each box contains the scores of all iterations.

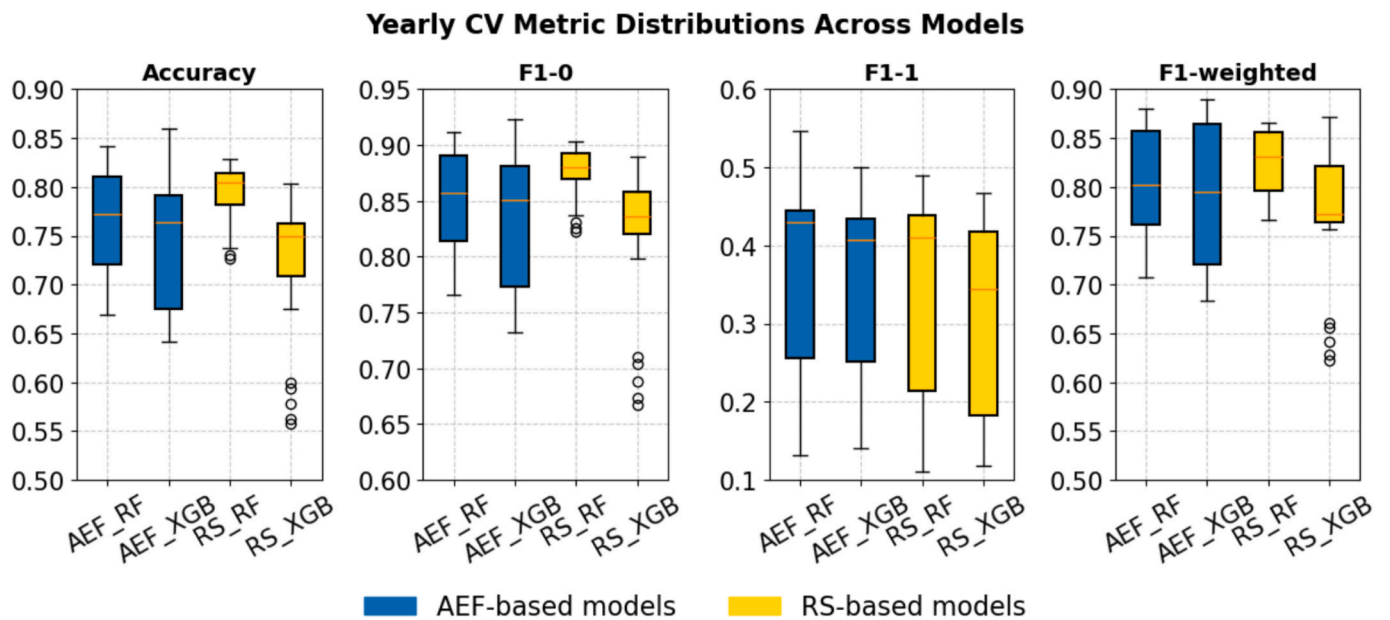


Fig. 7. The field-level cover crop mapping model performance (accuracy, F1-0, F1-1, and F1-weighted) under the yearly CV scheme. Each experiment is repeated five times, and each box contains the scores of all iterations.

models generally are competitive with RS-based models, especially in yield prediction. For example, under the state-year CV for crop yield prediction, AEF-based XGB model achieved the R^2 values of 0.78 for corn, soybean, and winter wheat, compared to 0.80, 0.79, and 0.71 for the RS-based XGB model.

Better spatial coverage: We also noticed that AEF embedding dataset has better spatial coverage than commonly used RS imagery, as it provides consistent, gap-free representations across space and time and is not subject to cloud contamination. For instance, there are a total number of 90,210 corn yield records in the Corteva yield dataset. We successfully downloaded AEF embeddings for all but downloaded RS features for 89,938 of them. The missing RS records are mostly due to data gap in Landsat or insufficient number of time-series observations

for harmonic fitting.

5.2. Limitations of AEF embeddings

5.2.1. Low interpretability

One major limitation when implementing AEF embeddings is the low interpretability. Each individual band of the 64-dimension annual embeddings is labeled sequentially from A00 to A63, without clear physical meanings. An example of feature importance analysis is presented in Fig. 8, in which the feature importance in FM-based and RS-based RF models was quantified by the mean decrease in impurity across all trees in the forest (Breiman, 2001). Based on feature importance identified by RS-based RF models (Fig. 8 (b)), it is evident that satellite-derived

Table 6

Space-transfer evaluation results between the East and the West ecoregions in the U.S. Corn Belt for each agricultural downstream task. The best performer is highlighted in bold for each case.

Task	Scale	Crop	Scheme	AEF-based RF		RS-based RF	
				R ²	RMSE	R ²	RMSE
Yield Prediction	County	Corn	East → West	0.02	2.71	0.66	1.61
			West → East	0.51	1.23	0.60	1.12
		Soybean	East → West	−0.28	0.94	0.53	0.56
			West → East	0.39	0.44	0.59	0.37
	Field	Corn	East → West	0.25	2.65	0.38	2.41
			West → East	0.33	2.48	0.34	2.45
		Soybean	East → West	0.08	0.99	0.19	0.93
			West → East	0.23	1.00	0.28	0.97
Tillage Mapping	County	N/A		R ²	RMSE	R ²	RMSE
			East → West	−0.35	0.21	0.11	0.17
	Field	N/A	West → East	0.43	0.14	0.21	0.15
				Accuracy	F1-weighted	Accuracy	F1-weighted
			East → West	0.64	0.64	0.67	0.68
			West → East	0.66	0.67	0.72	0.72
Cover Crop Mapping	Field	N/A		Accuracy	F1-weighted	Accuracy	F1-weighted
			East → West	0.88	0.87	0.84	0.85
			West → East	0.48	0.53	0.52	0.57

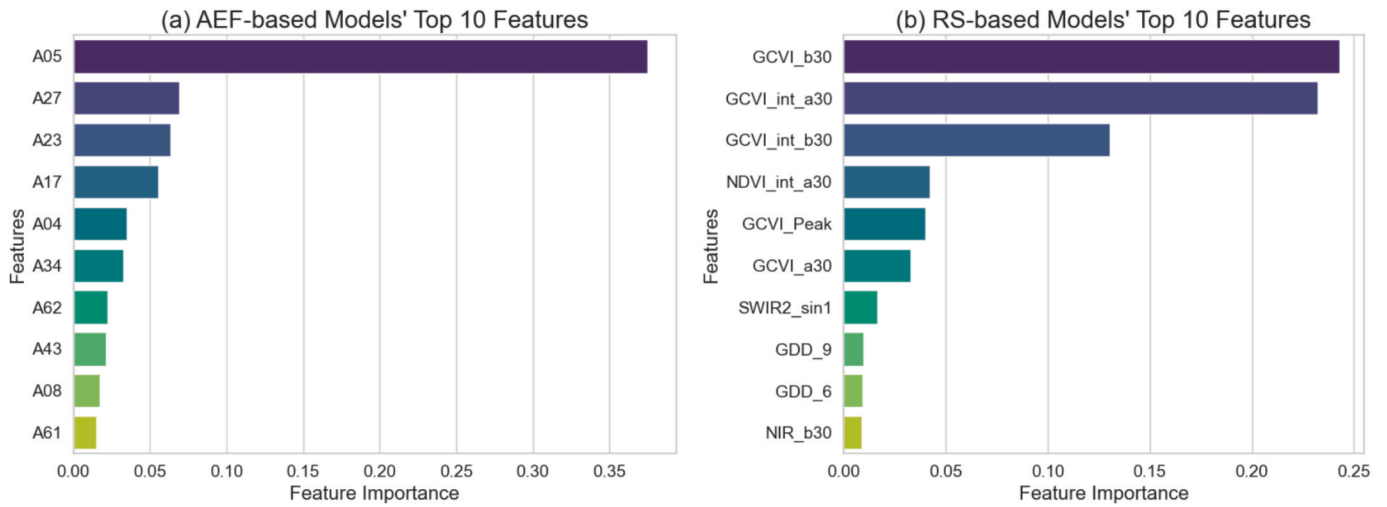


Fig. 8. The Top 10 important features in (a) AEF-based RF models and (b) RS-based RF models for county-level corn yield prediction.

features contribute more than climate variables, with GCVI proving more informative than NDVI and raw spectral bands. This information

can guide feature selection and model design. Meanwhile, A05 is the most important feature in FM-based RF models and surpasses the second

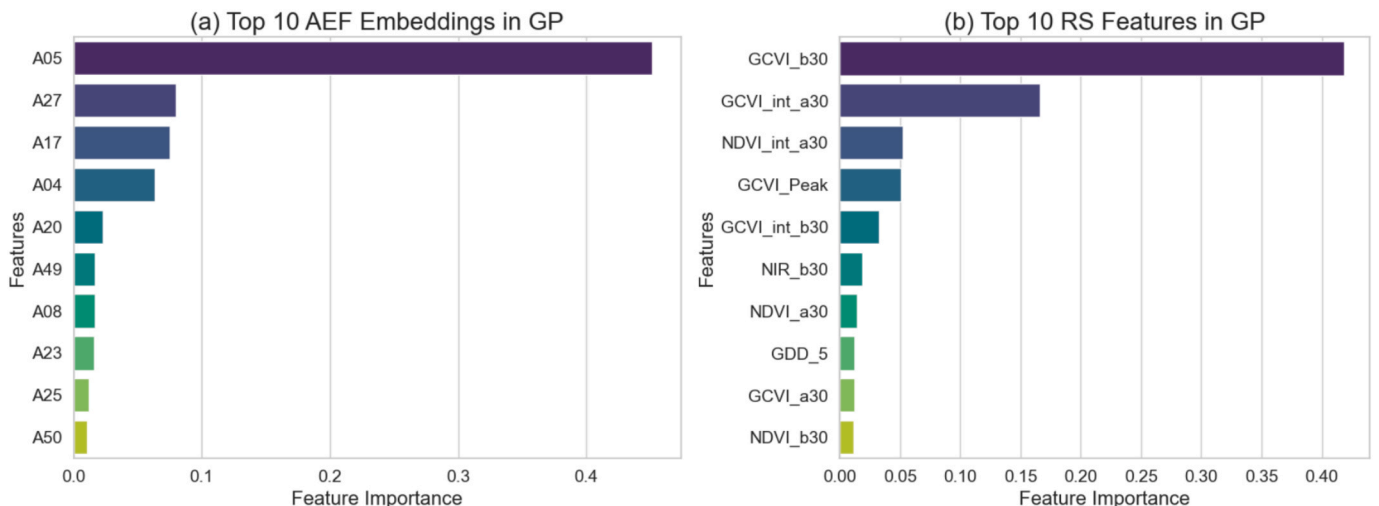


Fig. 9. The Top 10 important features in (a) AEF-based RF models and (b) RS-based RF models for county-level corn yield prediction in GP.

A17 by a large margin (Fig. 8 (a)). However, it is uncertain why A05 is the top contributor to corn yield prediction, given that its meaning is unknown. Correlation analysis shows that A05 is negatively correlated with GCVI features and positively correlated with GDD features (Fig. S2), suggesting that A05 captures a mix of crop greenness and temperature signals. However, this relationship does not appear stable, as for instance the most important AEF features for soybean yields differed greatly from those for corn, even though the RS-based models had similar feature importance for the two crops (Fig. S3).

5.2.2. Limited spatial transferability

AEF embeddings exhibit more pronounced geographic shifts than remote sensing features in certain tasks. Our space-transfer experiments show that AEF-based models consistently underperformed RS-based models in yield prediction (Table 6). We visualized the feature importance ranked by AEF-based and RS-based RF models for county-level corn yield prediction in GP (Fig. 9) and ETF (Fig. 10). Feature importance analysis indicates that the AEF-based models trained locally in GP and ETF prioritize different features, with A05 ranking highest in GP (Fig. 9 (a)) but A23 emerging as the top contributor in ETF (Fig. 10 (a)). While A05 and A23 are among the three most important embeddings in the overall U.S. RF model (Fig. 8 (a)), region-specific models emphasize different features.

More importantly, correlation analysis shows that the same embedding bands emphasize different signals across regions (Table S1). A05 is strongly correlated with both GCVI_int_a30 and GCVI_int_b30 in GP, whereas its correlation coefficients with both GCVI features are less than 0.18 in ETF. As a result, the AEF-based RF model trained in GP, which relies heavily on A05, fails to capture GCVI-related signals when transferred to ETF, leading to poor yield prediction performance. A similar pattern occurs when training in ETF and testing in GP. Meanwhile, t-SNE visualizations of the AEF embeddings indicate substantial distribution shifts in the feature space between ETF and GP (Fig. S4).

These geographic shifts can be even more pronounced across countries or continents. To confirm this, we further conducted space-transfer experiments from the U.S. to Argentina for county-level soybean yield prediction in 2019–2024. The unit of yield was converted to tonnes per hectare in both countries. Notably, AEF-based models trained in the U.S. significantly underperformed the RS-based models and failed to make accurate yield predictions in Argentina across all years (Table 7). These analyses indicate that AEF embeddings, even for the same band, encode and prioritize signals unevenly across different regions. Consequently, AEF-based models trained in one region learn region-specific association between the embeddings and the yield information, which can hinder

transfer to the other region.

The limited spatial transferability may arise from multiple contributing factors. During model training, AEF embeddings were decoded to not only reconstruct satellite images but also DEM and gravity fields, which are region-specific and relatively static, and may limit the embeddings to capturing localized information. Another possible reason is that geolocated articles from Wikipedia were used to provide text-based information as inputs to a text encoder, which were updated via contrastive learning with vision-based models. Such text-based information might be region-specific, resulting in distinct embeddings across space, particularly when comparing regions across countries or continents. Verifications of these speculations require ablation experiments based on the AEF model architecture and pre-trained weights, which are not publicly available.

5.2.3. Other limitations

In addition, we observed several other limitations in AEF embeddings. One is that AEF-based models showed temporal variations under yearly CV similar to RS-based models. This finding suggests that although AEF was trained to realize time-continuous featurization, it is still subject to the availability and quality of EO, leading to featurization with varying performance across different years. Another limitation is that AEF lacks temporal sensitivity and therefore cannot support time-critical tasks such as in-season yield prediction. The annual AEF embeddings are derived from year-long time series of Earth observations and therefore are not available until the following year. In other words, the current AEF embedding dataset may only be suitable for historical retrospective analysis but cannot be used for real-time analyses. Furthermore, the AEF model architecture and pre-trained weights are not publicly accessible. This restricts users' flexibility to reproduce embeddings or analyze their physical interpretability.

6. Conclusion

The rapid development of GFMs shows promises to overcome the long-standing limitations in EO. Google DeepMind's AEF is the first GFM that supports multi-modal EO featurization across continuous time, and the global, annual, and analysis-ready AEF embeddings are available from 2017 to 2024 on GEE. Internal evaluations show that AEF embeddings outperform existing featurization approaches in multiple EO tasks. However, those tasks are mainly about land cover and land use classification, with no systematic evaluation on its performance in agricultural tasks. This study comprehensively evaluated AEF embeddings on three critical agricultural downstream tasks at both regional

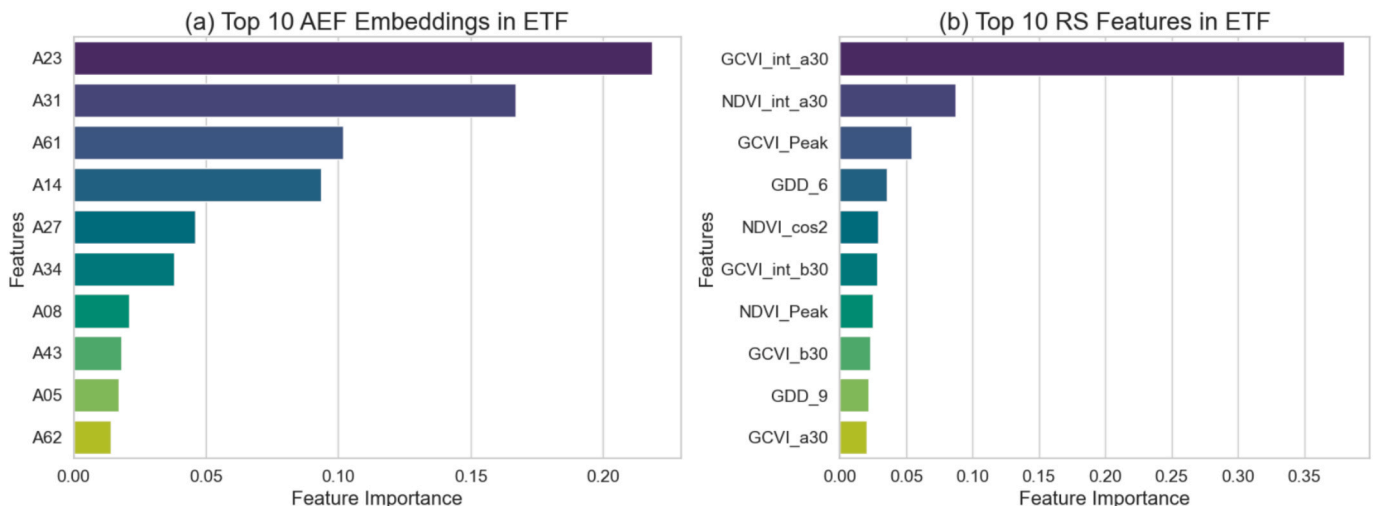


Fig. 10. The Top 10 important features in (a) AEF-based RF models and (b) RS-based RF models for county-level corn yield prediction in ETF.

Table 7
Space-transfer evaluation results of county-level soybean yield prediction in Argentina.

Scheme	Year	AEF				RS			
		RF		XGB		RF		XGB	
		R ²	RMSE	R ²	RMSE	R ²	RMSE	R ²	RMSE
US → Argentina	2019	−5.24	1.84	−5.67	1.62	0.04	0.72	0.05	0.72
	2020	−3.09	1.54	−3.53	1.43	−0.16	0.81	−0.14	0.81
	2021	−3.04	1.35	−3.54	1.40	−0.29	0.75	−0.30	0.76
	2022	−3.08	1.33	−3.58	0.67	0.01	0.65	−0.02	0.66
	2023	−0.64	0.64	−0.82	1.59	−0.04	0.53	−0.06	0.53
	2024	−3.84	1.51	−4.35	1.49	0.30	0.58	0.27	0.59
	All	−2.13	1.42	−2.45	1.90	0.27	0.68	0.26	0.69

and field levels in the U.S., including crop yield prediction, tillage mapping, and cover crop mapping. We have introduced and demonstrated an evaluation framework that can serve as a methodological baseline for future GFM assessments, emphasizing the need for evaluation of not only local performance, but also spatial, temporal, and scale transferability.

Evaluation results show that AEF-based models have strong predictive power and are competitive with RS-based models in several scenarios. AEF also has better spatial coverage and requires minimal preprocessing before modeling. It has the potential to transform the current EO-based agricultural research. However, we also identified several limitations in AEF, including limited spatial transferability and low interpretability. In particular, AEF-based models exhibited significant geographic shifts and failed to generate reasonable yield predictions across regions and countries. We also found it is challenging to interpret feature importance in AEF-based models, since there is no clear physical meaning in each band of the embeddings. Beyond that, other limitations in AEF embeddings include temporal variability and limited time sensitivity. Collectively, these limitations raise concerns when applying AEF embeddings in agriculture, where time sensitivity, generalizability, and interpretability is critical. Future AEF versions or other GFM development can tackle these limitations through generating monthly/seasonal embeddings, balancing training site selection, and incorporating feature attribution techniques to enhance interpretability.

CRediT authorship contribution statement

Yuchi Ma: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yawen Shen:** Writing – review & editing, Data curation. **Anu Swatantran:** Writing – review & editing, Conceptualization. **David B. Lobell:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Yuchi Ma reports financial support was provided by Google LLC. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.]

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expressed in this study are those of the authors and do not necessarily reflect the views of NASA or Google.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jag.2026.105258>.

Data availability

Data and codes for county-level yield prediction are available on Github (https://github.com/yuchima8/Harvest_AlphaEarth). Other data can be made available on request.

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